Notes 01/29

Def. A positive natural number is a nonzero element of $\mathbb{N}$. We denote the positive natural #s by $\mathbb{N}^+$.

In other words, $\mathbb{N}^+ = \mathbb{N} - \{0\}$
$\overset{\text{def}}{=} \mathbb{N} \setminus \{0\}$

Cor 4 If $n \in \mathbb{N}^+$, then n has a unique predecessor in $\mathbb{N}$

Proof. By Thm. 2 n has a predecessor in $\mathbb{N}$. To see uniqueness, suppose $\sigma(a) = \sigma(n)$ with $a, b \in \mathbb{N}$

By Axiom 3, σ is injective. Thus $\sigma(a) = \sigma(b) \Rightarrow a = b$

$\boxed{\text{Since sigma } a = \text{sigma } b, \text{ b and a are =}}$

Define a predecessor function $\pi : \mathbb{N}^+ \to \mathbb{N}$ as follows. If $n \in \mathbb{N}^+$, Cor 4. shows there is a unique $a \in \mathbb{N}$ such that $\sigma(a) = n$, we define $\pi(a)$ to be a.
I.e. $\pi(n) \overset{\text{def}}{=} a$ where $\sigma(a) = n$

x:

Homework: define a restricted succesor function σ' and show σ' & π are inverses

Axiom 5 (Iteration): Let $f: S \to S$ be a function. The nth iteration of f is a function from S to itself. $\underbrace{f \circ f \circ f \cdots}_{f^n} = f^n : S \to S$

Such functions satisfy the following:
(i) f^0 is the identity function on S
(ii) $f^{\sigma(n)} = f \circ f^n$

Exercise 12

Prove that $f' = f$

$$f' = f^{\sigma(0)} = f \circ f^{0} = f.$$

$1 = \sigma(0)$ Axiom S(ii) Axiom S(i)

Prove $f^2 = f \circ f$

$f^2 = f^{\sigma(1)} = f^{\sigma(1)} = f \circ f'$ ✓

Addition

Def. Let $m, n \in \mathbb{N}$ Recall $\sigma^n : \mathbb{N} \to \mathbb{N}$ is the nth iteration of the successor m

$m+n$ to be the $\sigma^n(m)$ I.e. $m+n \overset{def}{=} \sigma^n(m)$

Intuition $2+3 \overset{def}{=} \sigma^3(2) = 2+1+1 = 5$

We see that addition defines a function from $\mathbb{N} \times \mathbb{N} \to \mathbb{N}$ and

Such function are called binary operation

Theorem 4: For all $m \in \mathbb{N}$, one has $m+0 = m$

Proof: We compute $m+0 = \sigma^0(m) = m$

Lemma $\forall\, m, n \in \mathbb{N}$, $m + \sigma(n) = \sigma(m+n)$

Theorem 6. For all $\forall\, m \in \mathbb{N}$, $m+1 = \sigma(m)$

Theorem 7 (Associative Law): For all $x, y, z \in \mathbb{N}$, $x+(y+z) = (x+y)+z$

Proof: We proceed by induction. Let $x, y \in \mathbb{N}$ be arbitrary

Define $S_{x,y} = \{z \in \mathbb{N} : x+(y+z) = (x+y)+z\}$

Firstly, $S_{x,y} \subset \mathbb{N}$. Consider we compute $x+(y+0) = x+y = (x+y) = (x+y)+0$

[Thm 4] [Considering] [By Th 4]

We see $0 \in S_{x,y}$

We see $\emptyset \in S_{x,y}$ ... Now suppose $n \in S_{x,y}$

We have $x+(y+\sigma(n)) = x+\sigma(y+n)$ [Lemma 5]

$= \sigma(x+(y+n))$ [Lemma 5]

$= \sigma((x+y)+n)$ [since $n \in S_{x,y}$]

$= (x+y)+\sigma(n)$ [Lemma 5]

We see $\sigma(n) \in S_{x,y}$ Thus $\forall x\ \forall y$ (Induction) shows $S_{x,y} = \mathbb{N}$. Since x, y are arbitrary, wherefore

$x+(y+z) = (x+y)+z \quad \forall x,y,z \in \mathbb{N}$